LLM reads, code verifies

Can you trust what an LLM reads? When should it call the cloud? What if the data can't leave?

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Portfolio & resume

How it works & code

FROM PRODUCTION

90%+ less reconciliation time · sole engineer · 729 PRs

7 LLMs benchmarked on a hand-built 90-receipt, 8-currency set

87 s → 10 s worst-case latency with Gemini → Claude failover

Rolled back confidence-based auto-approval: why this harness exists

1 Can you trust what an LLM reads?

Not its confidence. Check every answer with code.

FOUR CHECKS ON EVERY MODEL ANSWER

- **Sanitizer:** injection patterns in EN/KO/JA/ZH, field whitelist
- **Arithmetic:** subtotal + tax + tip + discount = total
- **Date plausibility**
- **Two-reader agreement:** the total and date must appear in the OCR text

WITH NO ANSWER KEY, THE CHECKS CAUGHT

38 of 51
wrong answers from the local model

5 of 8
wrong answers from the cloud model

43 adversarial receipts: 16 prompt injections, 0 fooled; 21 date and arithmetic traps, all caught; 0 leaked.

2 When should it call the cloud?

Local first, cloud when the checks fail, people last.

ON THIS MACHINE

T1 · local OCR + rules
RapidOCR + Tesseract
73% · 55%
4.7 s · $0 · CPU

T2 · local VLM
Qwen2.5-VL 3B
83% · 81%
28 s · $0 · 4 GB GPU

CLOUD

T3 · cloud, cheap
Gemini 3.1 Flash-Lite
97% · 87%
1.8 s · $0.54/1k

T4 · cloud, strong
Gemini 3.6 Flash
98% · 87%
1.9 s · $1.42/1k · Claude failover

Each reader alone: share of receipts with date and total both exact, **scans · photos**; median time and cost.

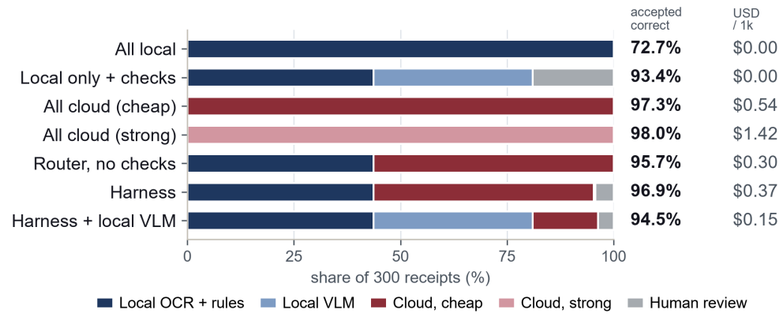


Fig A. Where 300 scans end up under each policy, accepted-answer accuracy and USD per 1,000.

The harness: **96.9% correct** with 56% of receipts sent to the cloud and 4.3% to people, $0.37 per 1,000 (all cloud: 97.3%, $0.54). The strong model adds 0.7 points for 2.7× the price: **escalate to people, not to a bigger model.**

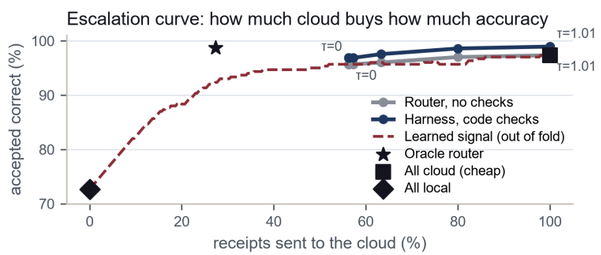


Fig B. Escalation curve: share sent to the cloud vs accuracy as the threshold moves; ★ = a router that knows the answers.

A perfect router needs only 27% cloud for 98.7%; the rule signal needs 56% for 95.7%. **The bottleneck is the escalation signal, not the model.**

A learned signal (logistic regression, 14 features, out-of-fold AUC 0.89) did no better at the same cloud share, so the two rules stayed.

3 What if the data can't leave?

Run local-only: same checks, more human review.

	CLOUD HARNESS	LOCAL ONLY
Scans	**96.9%** correct 4.3% to people · $0.37/1k	**93.4%** correct 19% to people · $0 · 27 s
Photos	**88.5%** correct 21.5% to people	**89.1%** correct 31.5% to people

On phone photos the two modes match on accuracy and differ only in review load; on scans local only gives up 3.4 points. **Keeping data in costs mostly people's time.**

In cloud mode the image leaves the machine at one step only; every check and routing decision runs locally.

Check rules are domain assets: a rule built from scan errors caught 10 of 16 there with no false alarms, but 1 receipt on held-out photos, with 1 false alarm.

NEXT

10k receipts; invoices and statements; layout features as escalation signals; the oracle gap as a weekly drift metric.

300 scanned receipts (ICDAR 2019 SROIE) · 200 phone photos (WildReceipt) · 43 synthetic adversarial receipts. Date and total must match exactly; scoring is code, never a model.

Run of 2026-10-31 · code, manifests and every result are in the repository

Sources. ICDAR 2019 SROIE (Huang et al., arXiv:2103.10213), CC-BY-4.0 · WildReceipt (Sun et al., arXiv:2103.14470) · Tesseract 5.4, RapidOCR / PP-OCR, Qwen2.5-VL (Apache 2.0) via Ollama · Google Gemini API pricing page, 2026-10-31 · Fallback chain and sanitizer adapted from the author's open-source boda (MIT). Production figures are from the author's 2026 engagement; the platform is private.